4.3 Reel Core Shaft Support

The K5 Expert machine is designed to process the substrate from 154mm inside core, 152mm (6 inch) outside diameter.

A reel shaft is used to support the substrate when it is being wound from cardboard, plastic or metallic cores.

The shaft has a diameter of 152mm and is manufactured to fit inside a core with a minimum inside diameter of 154mm.

The reel shaft is supported by the machine chucks when the substrate is loaded on to the machine.

Bearings are fitted to the shafts to allow the shafts to be supported from the machine side-plates.

Figure 4.3-1 Example of a roll of film on a reel shaft

Warning!

Make sure that the appropriate equipment is used to handle and support the reel shafts when changing between substrate rolls.

DANGER!

Static electricity can build up in the substrate rolls. There is a risk of electrical shock from static discharge from substrate when changing rolls.

Centralising the roll on the shaft

The winding mechanism is designed to process the roll in the centre of it.

The shaft should be placed in the roll of substrate, and a measurement taken from the edge of the roll to the centre of the support bearing. The measurements on the left and the right sides have to be equal, to ensure the roll is in the centre.

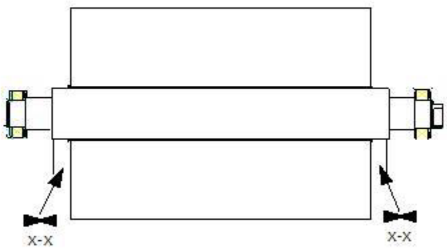

Figure 4.3-2 Location of the roll on a shaft

	Note !
	If the roll is not in the centre of the winding rollers, the inline monitor system will not operate correctly

	Warning !
	Do not operate the machine outside the limits given in the specification. The minimum width should not be exceeded.

Cardboard Core Quality

If the core is manufactured from poor quality products, it can affect the winding characteristics of the machine. It is recommended that cardboard cores are spirally wound cores, trimmed finish with an inside diameter of 153.12mm with a wall thickness of 14mm.

4.4 Operation of the Reel Shafts

It is recommended that the paper core wall thickness should be a minimum of 14-17mm thick to stop potential winding issues with deformation.

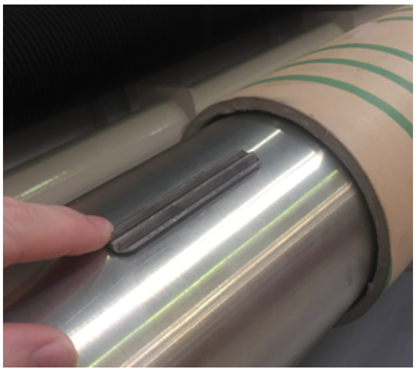

Figure 4.4-1 Reel shaft inserted into a cardboard core

The roll or core (shell) should be placed centrally on the reel shaft using a loading and centring device, if available. To check that it is central, a quick measurement from the edge of the reel to the inside of the bearings will suffice.

To engage the reel shaft, insert the T-wrench supplied into the end of the shaft and turn it clockwise until the core is locked to the shaft (Approx. 100 Nm Torque). The rotation on the end of the shaft results in the extending of the locking segments from within the shaft.

Alternatively, an air wrench can be used to tighten the shaft within the roll.

	Note !
	If the reel shaft is not tightened sufficiently, slippage between the core and shaft may take place. This can lead to telescoping of rolls.

To disengage the reel shaft from the core, turn the T wrench in an anti-clockwise direction, until the segments are retracted.

Figure 4.4-2 Expanding a mechanical reel shaft within a roll

	Caution !
	Do not use excessive tensioning torque on the reel shaft as mechanical damage can take place. This will lead to a shortened life span of the equipment.

	**Note !**
	It has been found that when moving a previously used empty core from the unwind to rewind position; it should be retightened as the core may now be loose, as it is not compressed by the material that was on it.

4.5 Threading the Substrate through the Winding System

Refer to the machine web path diagram to ensure that the correct web path is followed through the machine.

Before any rolls are placed in the winding mechanism, it is suggested that a leader tape be threaded through the machine, to assist the threading procedure.

However, it is normal practice to load a new unwind roll into the machine before a finished rewind roll is removed. This allows the leading edge of a new unwind roll to be attached to the tail end of a previous roll to assist in the threading procedure.

It is also normal practice to cut the substrate, on the unwind roll, and therefore provide a lead edge to ease its path through the winding mechanism.

Warning !

Ensure that the movement isolator is in the off position and locked out before entering the winding mechanism area.

Under no circumstances, override any interlock, either mechanically or electrically

It is suggested that thin double-sided self-adhesive tape must be used on all splices and to fix the web to the rewind core, taking care to avoid any creasing of the substrate.

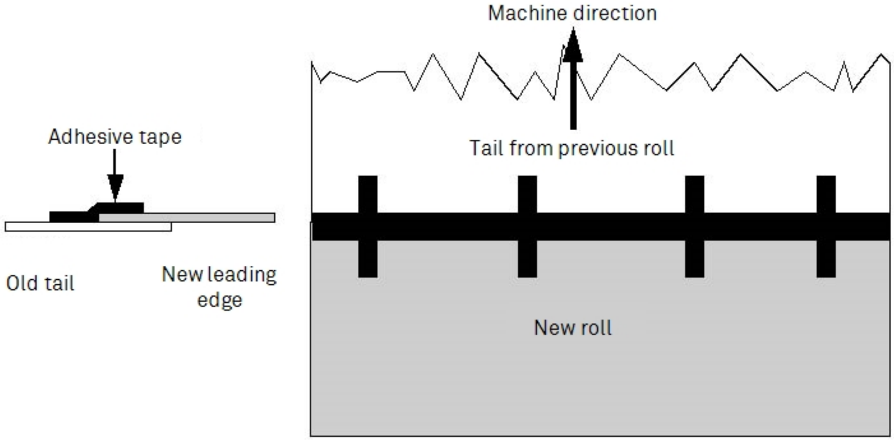

Figure 4.5-1 Examples of Joining of the Substrate at the Unwind Roll

Any creases located in the substrate at splices or when attaching it to the core will affect the quality of the winding.

Substrate Details

When threading is completed, the following substrate parameters are required by the machine drive system to allow normal operation:-

- Substrate Thickness
- Substrate Width
- Substrate Type

This allows the operator to input the initial start conditions of the drive system in the HMI. Insert this information into the computer as described later in this manual.

	Note !
	Ensure that all adhesive tape used for threading the machine is removed before running the drive system. Any adhesive tape remaining in the winding mechanism can lead to web breaks etc.

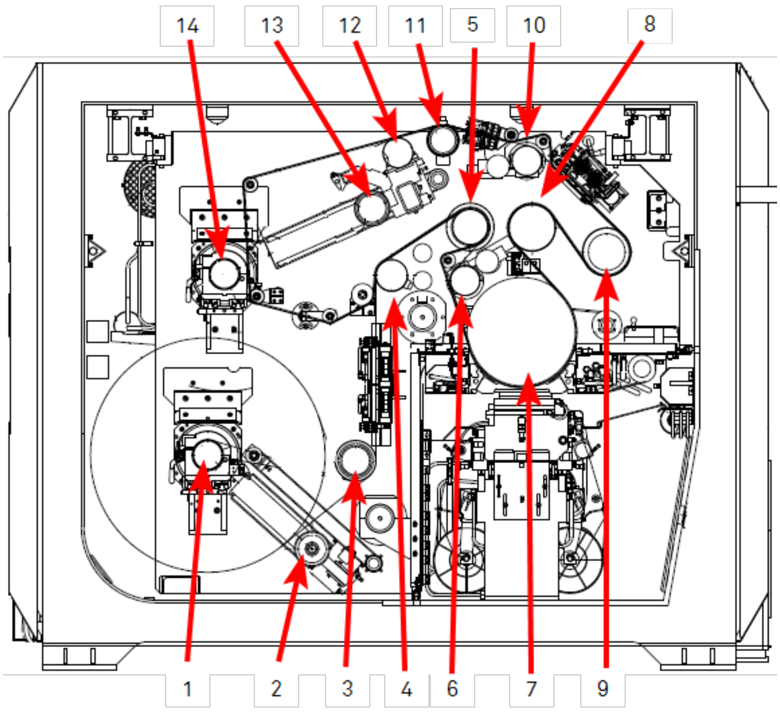

Figure 4.5-2 Web Path Drawing

Roller Number	Description
1	Unwind with shafted core - Driven
2	Unwind Layon Roller
3	Spreader Roller
4	Load cell Roller
5	Cork Covered Roller - Driven
6	Bowed Roller
7	Water Cooled Drum – Driven
8	Water Cooled Roller – Driven
9	Cork Covered Draw Roller – Driven
10	Bow Spreader Roller Cork Covered
11	Load cell Roller
12	Bow Spreader Roller Cork Covered

Table 4.5-1 Roller Description

Roller Number	Description
13	Rewind Layon Roller
14	Rewind with shafted core -Driven

Table 4.5-1 Roller Description

4.6 Direction Selection of Winder Drives

The unwind and rewind rolls can rotate in either a clockwise or counter-clockwise direction. This allows the web to be threaded to and from the rewind and unwind rolls using an underwind or overwind method.

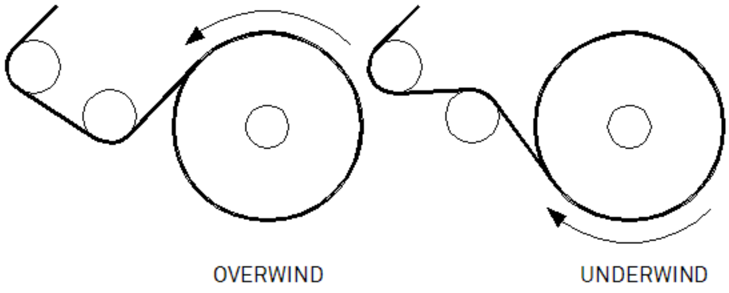

Figure 4.6-1 Diagram Showing Underwind/Overwind Web Path from a Roll (Illustration Only)

This is provided to:

- allow the coating of the inside or outside surfaces of the roll
- wind the finished product with the coating on the inside or outside of the roll

Note !

At the unwind roll, the side of the substrate that will be coated is the side facing down to the floor.

This should be considered in regards to inside/outside treated substrates

Use the HMI to select the winding modes for the unwind and rewind rolls. This is set in the RECIPES WINDING screen. It can be also modified in the CONFIGURATION screen, if the substrate is not under tension.

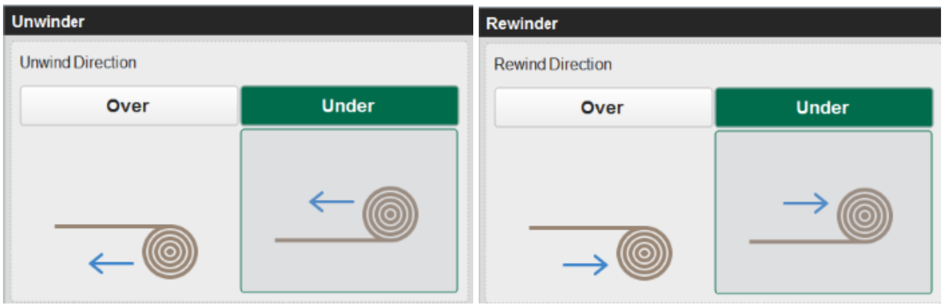

Figure 4.6-2 Screen captures of direction selection interfaces on RECIPES HMI screens

Touching the button, will highlight the mode, as it will be then coloured. The above screens show the winders selected for underwind direction.

Note !

To change this selection, the drive system must be in a powered down state (that is, tension off).

4.7 How to Measure the Roll Diameters

The winding mechanism drive system requires the unwind roll and rewind core diameters to determine the initial speed that the winder motors need to rotate.

This is achieved by moving the unwind and rewind layarms into contact with the roll and core.

Before this, the reset layarms sequence must be carried out to find the home position (limit switch) for the layarms.

Warning !
Ensure the area is clear before pressing the GET DIAMETER button.
The visible and audible alarms will be active while the layarms are moving.

To measure the roll diameter:

1. Make sure that the web is not running
2. Make sure that the web area is clear
3. Go to the rear HMI
4. Press Reset Layarms to start the reset layarms sequence
5. Wait until the reset layarms sequenc is complete
6. Press the get diameter button

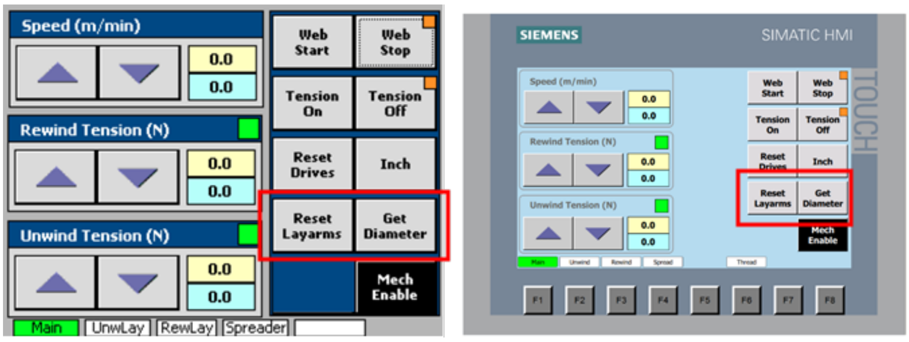

Figure 4.7-1 Example of the Reset Layarms and Get Diameter buttons on the rear HMI screen